



SE4020 – Mobile Application Design and Development

Assignment 02 – NurseryConnect

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1 Introduction

This report documents the design, implementation, testing, and evaluation of the NurseryConnect application developed for SE4020 - Mobile Application Design and Development. The submission addresses both Part A and Part B of Assignment 2. Part A extends the Assignment 1 NurseryConnect iOS application into a fully featured iPadOS application built with SwiftUI, incorporating platform-native features, advanced Apple frameworks, and complete analytics and secure messaging systems. Part B presents a visionOS spatial computing companion (Option 02) that reimagines the keyworker's childcare management environment as an immersive spatial workspace using RealityKit and Apple Vision Pro.

NurseryConnect serves the daily operational needs of keyworkers at Little Stars Nursery and Daycare, supporting workflows across four core domains: daily diary observation logging, real-time attendance management, structured incident reporting, and secure parent-keyworker messaging. The application is built on SwiftUI and Core Data, follows the MVVM architectural pattern with a coordinator layer for iPad state sharing, and maintains full compliance with GDPR, EYFS statutory requirements, and ICO guidance throughout every layer of the data model.

2 Part A – iPadOS Application

2.1 Extension from Assignment 1

This section explicitly identifies what has been carried forward from Assignment 1 and what constitutes new development introduced in Assignment 2. The 22nd April 2026 GitHub commit serves as the Assignment 1 baseline.

2.1.1 Features Carried Forward from Assignment 1

The following modules were developed as part of Assignment 1 and have been carried forward, in several cases with targeted enhancements as described in Section 2.1.2:

- Incident Reporting module: IncidentListView, IncidentComposerOverlay, IncidentDetailView, and BodyMapView. Carried forward and significantly enhanced in Assignment 2 with urgency banners, compliance tiles, segmented scope filters, and a SafeAreaInset floating action button.
- Daily Diary module: DiaryTimelineView and DailyDiaryListView with entry types covering activities, sleep, meals, nappies, wellbeing ratings, and milestone photos. Enhanced in Assignment 2 with chart-driven entry highlighting and an external add-sheet binding for coordinator control.
- FaceBlurEditorView: A Vision-framework face detection and Core Image blurring pipeline applied to milestone photos for GDPR compliance. Refactored in Assignment 2 with improved EXIF orientation normalization and a 1600 pt image cap to reduce memory pressure.
- Core Data model: Child, AttendanceRecord, DiaryEntry, DiaryEntryCorrection, and Incident entities with all sync fields. Extended in Assignment 2 with the Message and MessageThread entities.
- AppTheme design system: Migrated from explicit size fonts to Dynamic Type-compliant semantic styles in Assignment 2.
- AttendanceView and KeyworkerDashboardView: Carried forward from Assignment 1; the dashboard's absent section was given independent collapsibility in Assignment 2.

2.1.2 New Features Introduced in Assignment 2

The following modules and views are entirely new to Assignment 2 and represent the substantive sprint-two development effort:

Table 1: New Features Introduced in Assignment 2

Feature	Key Files	Description
Analytics Dashboard	ChildAnalyticsDashboardView, MoodTrendChart, WeeklyActivityDistributionChart, ChildMonthlyAttendanceChart, MessageActivityChart, AnalyticsDataService	Full analytics layer built on Swift Charts, GDPR-gated by keyworker assignment
Secure Messaging System	MessagingViewModel, ThreadDetailView, MessageInboxView, EndOfDaySummarySheet, MessagingGDPRScope	Core Data-backed inbox with three-tier grouping, chat bubble threads, reply restrictions, and end-of-day summary with rate limiting
iPad Split-View Shell	KeyworkerIPadShellView, KeyworkerIPadCoordinator, ChildrenSidebarView, AdaptiveRootView	NavigationSplitView-based iPad workspace with 62 pt minimized avatar rail
Child Journal View	ChildJournalView, ChildJournalSegmentBar, ChildDetailPanelView	Split-view trailing panel showing Daily Diary and Charts segments switchable by swipe or capsule segment bar
Dashboard Quick Check-In	DashboardQuickCheckInSheet	Compact sheet (.medium detent) for rapid attendance check-in without navigation
Milestone Gallery View	MilestoneEntriesView	Gallery-style view for milestone photos with blurred-face count and EYFS development note
Keyboard Shortcuts	KeyworkerCommands.swift	CommandMenu shortcuts for iPad external keyboard: New Diary Entry, New Incident, Jump to Messages, Dismiss Sheet
Journal Summary Card	JournalAnalyticsSummaryCard	Compact summary card showing today's mood average, entry count, and message activity sparkline

2.2 Platform-Specific iPadOS Features

The application incorporates five distinct iPadOS-native capabilities that fundamentally differentiate the iPad experience from the iPhone companion. These features leverage platform affordances that are unavailable or inappropriate on iPhone.

2.2.2 NavigationSplitView and Multi-Column Navigation

The primary iPadOS feature is the NavigationSplitView-based KeyworkerIPadShellView, which delivers the multi-column workspace expected of professional iPadOS productivity applications. The sidebar column (240–320 pt) presents a searchable and sortable children list with compact ChildSidebarRowView cards showing attendance status badges, diary-completion dots, and no-photography warning indicators. The detail panel (400 pt) renders the selected child's full journal workspace. When the sidebar is collapsed, a minimized rail of 62 pt child avatar tiles remains visible for rapid child-switching without reopening the full sidebar, a pattern consistent with Apple's recommended compact sidebar design demonstrated in WWDC24 Session 10147 [1].

```
NavigationSplitView(columnVisibility: $splitColumnVisibility) {  
    ChildrenSidebarView(...)  
    .navigationSplitViewColumnWidth(min: 240, ideal: 280, max: 320)  
} detail: {  
    KeyworkerIPadDetailContent(...)  
}
```

The AdaptiveRootView reads @Environment(\.horizontalSizeClass) and routes to KeyworkerIPadShellView on .regular (iPad) and KeyworkerDashboardView on .compact (iPhone), providing correct platform behavior from a single shared codebase.

2.2.3 External Keyboard Shortcuts

The KeyworkerCommands.swift file registers a CommandMenu that provides four keyboard shortcuts accessible when an external keyboard is connected to the iPad. These shortcuts appear in the system menu bar on iPadOS and follow the convention of pro-user productivity applications.

Table 2: Keyboard Shortcuts Registered in KeyworkerCommands.swift

Keyboard Shortcut	Action	Context
⌘ N	New Diary Entry	Opens the diary entry composer sheet from any screen
⌘ I	New Incident	Opens the incident reporting overlay
⌘ M	Jump to Messages	Switches the content panel to the messaging inbox
Esc	Dismiss Sheet	Dismisses any currently active sheet or modal presentation

2.2.4 Hover Effect for Pointer and Trackpad Interaction

The .hoverEffect(.highlight) modifier is applied to every ChildSidebarRowView in the children sidebar. When the user moves the cursor over a child row using a connected Magic Keyboard Trackpad or external mouse, the row responds with a platform-standard highlight effect. This interaction is unavailable on iPhone and is part of the iPadOS pointer interaction model. Combined with compactDisplayName formatting (First Name plus Last Initial) for space efficiency, the sidebar delivers a professional experience for nursery staff who use iPads with keyboard and trackpad accessories.

2.2.5 Presentation Detents for Sheet Sizing

All sheets in the iPadOS application use `.presentationDetents([.medium, .large])` in combination with `.presentationDragIndicator(visible)` and `.presentationCornerRadius(20)`. This provides the standard drag-to-resize behavior expected on a large iPad screen. The `DashboardQuickCheckInSheet` defaults to `.medium`, keeping the check-in form compact without obscuring the underlying attendance cards. The incident and message sheets default to `.large` to provide full working space.

2.2.6 Horizontal Size Class Adaptive Layout

Size-class checks via `@Environment(\.horizontalSizeClass)` appear in more than four files, driving substantive layout differences beyond the `AdaptiveRootView` split. These include the incident composer presentation style (`fullScreenCover` on iPhone, side overlay sharing coordinator state on iPad), analytics card height, and the incident detail sheet style. This approach follows Apple's adaptive layout pattern demonstrated in the *Develop in Swift Essentials* tutorial series [2].

2.3 Advanced Library Integration – Swift Charts

Swift Charts (Document & Rendering category, approved advanced library) is integrated across three analytics charts in the `ChildAnalyticsDashboardView`, providing data-driven welfare and attendance insights that directly support EYFS statutory obligations.

2.3.1 Analytics Dashboard Overview

The analytics dashboard is a completely new module in Assignment 2. It surfaces EYFS-relevant insights from the existing Core Data records, giving keyworkers a visual summary of each assigned child's wellbeing and attendance patterns without requiring manual review of individual diary entries. The dashboard is GDPR-gated: if the selected child is not assigned to the logged-in keyworker, a `ContentUnavailableView` renders an access-denied state. Each chart is a self-contained SwiftUI view with its own `@FetchRequest`, driven by `AnalyticsDataService`, a pure enum of static helper functions that transform Core Data rows into chart-ready value types, keeping the data transformation layer fully unit-testable.

2.3.2 Chart Implementations

Table 3: Swift Charts Implementations in the NurseryConnect Analytics Dashboard

Chart	Chart Type	Data Source	Key Regulatory Feature
Mood Trend	LineMark + RuleMark	Wellbeing diary entries, last 7 days	Red dashed welfare threshold at score 2, triggering EYFS welfare review banner
Weekly Activity Distribution	Tappable BarMark	All diary entry types, current week	Tap a bar to highlight matching diary entries in the journal timeline
Monthly Attendance	Custom LazyVGrid dot calendar	Attendance records, full current month	Green present, Red absent, Faded future days, Ring around today, full-month overview
Message Activity Sparkline	Compact BarMark	Parent messages, last 7 days	Embedded in JournalAnalyticsSummaryCard as a glanceable messaging activity indicator

The welfare threshold RuleMark in the Mood Trend chart is a regulatory implementation, not a cosmetic feature. The RuleMark API, introduced in WWDC23 Session 10037 [3], places the threshold line precisely in the chart's data coordinate system:

```
// MoodTrendChart.swift - EYFS welfare threshold line
RuleMark(y: .value("Threshold", 2))
    .foregroundColor(Color.red.opacity(0.7))
    .lineStyle(StrokeStyle(lineWidth: 1.5, dash: [5, 3]))
    .annotation(position: .trailing) {
        Text("Review").font(.caption2).foregroundColor(.red)
    }
```

The chartOverlay API informed the tap gesture implementation in WeeklyActivityDistributionChart, where tapping a bar reads the bar's data value from the chart's proxy and drives a highlight filter on the diary timeline.

2.3.3 Justification for Swift Charts over Third-Party Libraries

Swift Charts was chosen over all third-party charting alternatives for three reasons. First, as a first-party Apple framework, it participates fully in the accessibility stack: VoiceOver automatically describes each data point to screen reader users, Dynamic Type scales all annotation text with user font preferences, and Dark Mode colors adapt without code changes. Second, it has zero dependency overhead, no package manager complexity, no licensing concerns, and no binary size impact. Third, the RuleMark and chartOverlay APIs provide precisely the regulatory capabilities required: a data-anchored welfare threshold line and interactive bar selection for diary highlighting. These are genuine clinical features supporting EYFS welfare monitoring, not decorative additions.

2.4 Architecture and Code Quality

2.4.1 MVVM with Coordinators

Every feature in NurseryConnect follows a strict Model → ViewModel → View architecture. Models are Core Data NSManagedObject subclasses. ViewModels are @MainActor ObservableObject classes decorated with @Published properties, exposing processed, UI-ready state. Views read from ViewModels via @StateObject (for root ownership) and @ObservedObject (for injection), and contain no data-access or business logic code.

The KeyworkerIPadCoordinator is a shared @StateObject owned by AdaptiveRootView and injected as an @EnvironmentObject throughout the iPad view hierarchy. It maintains which incident or message thread is currently selected across the split-view columns, preventing state duplication between the sidebar and detail panels. This coordinator pattern cleanly separates per-feature logic (MessagingViewModel, IncidentViewModel) from cross-feature navigation state.

2.4.2 Core Data Entity Model

Table 4: NurseryConnect Core Data Entity Summary

Entity	Key Attributes	Relationships
Child	id (UUID), firstName, lastName, dateOfBirth, roomName, keyworkerName, photoConsent, authorisedCollectors (JSON)	attendanceRecords, diaryEntries, incidents, all cascades delete on child removal
AttendanceRecord	id, dayStart, checkInAt, checkOutAt, markedAbsent, droppedOffBy, collectedBy	child → Child (many-to-one)
DiaryEntry	id, timestamp, entryType, moodRating, milestonePhotoData (external storage), hasCorrections, isSubmittedToManager, syncState	child, corrections → DiaryEntryCorrection (cascade)
DiaryEntryCorrection	id, correctedAt, fieldName, oldValue, newValue, reason	entry → DiaryEntry (many-to-one)
Incident	id, timestamp, category, severity, status, isParentNotified, parentNotificationTimestamp, managerCountersigned, riddorRequired	child → Child (many-to-one)
MessageThread	id, childID, createdAt, initiatorRole, subject, isArchived	messages → Message (cascade)
Message	id, threadID, sentAt, body, senderRole, senderDisplayName, messageType, isRead	thread → MessageThread (many-to-one)

2.4.3 Offline-First Sync

All write operations have immediately persisted to the local Core Data store. A SyncQueueService runs every 90 seconds, on network reconnect (detected via NWPathMonitor), and on app foreground transition to push pending items to the backend. Every syncable entity carries syncState (synced / pending / failed), syncEnqueuedAt, syncAttemptCount, and lastSyncError. This design ensures keyworkers can record observations, check in children, and log incidents during Wi-Fi outages, with changes safely queued for upload on reconnection.

2.4.4 GDPR Scope Guards at Predicate Level

Data access is gated at the `NSFetchRequest` predicate level through `KeyworkerGDPRScope` and `MessagingGDPRScope` helper types. Rather than fetching all data and filtering in the `ViewModel`, which could expose unauthorized data through coding errors, the scope guards embed `keyworkerName` constraints directly into the predicate before the fetch is executed. Unauthorized child data never enters the `ViewModel` at all. All predicates are centralized in `AnalyticsDataService` as static functions and covered by dedicated unit tests to catch key-path typos at test time rather than as silent runtime crashes.

2.5 UI Design, Usability, and AI-Driven Mockup Process

The user interface was developed through an iterative AI-assisted design process. Three distinct UI mockup variations were generated using Claude Artifacts (`claude.ai`) as the primary tool, each evaluated against the keyworker role, the iPadOS platform conventions, and the childcare regulatory context before the final design was synthesized.

2.5.1 AI Tool Selection and Justification

Claude Artifacts was selected as the UI mockup generation tool for three primary reasons. First, it generates live interactive HTML and React prototypes that run in the browser, enabling evaluation of dynamic interactions such as sidebar collapse, sheet presentation, and tab switching, behaviors that static image mockups cannot demonstrate. Second, Claude Artifacts understands and correctly applies iPadOS-specific navigation patterns such as `NavigationSplitView` column layouts and sidebar rail behavior, which general-purpose image generators such as Midjourney do not. Third, the transition from Claude Artifacts prototype to SwiftUI implementation is direct: the component hierarchy, spacing decisions, and color values generated in the prototype can be translated to SwiftUI modifiers with minimal reinterpretation.

2.5.2 Generated Mockup Variations and Critical Evaluation

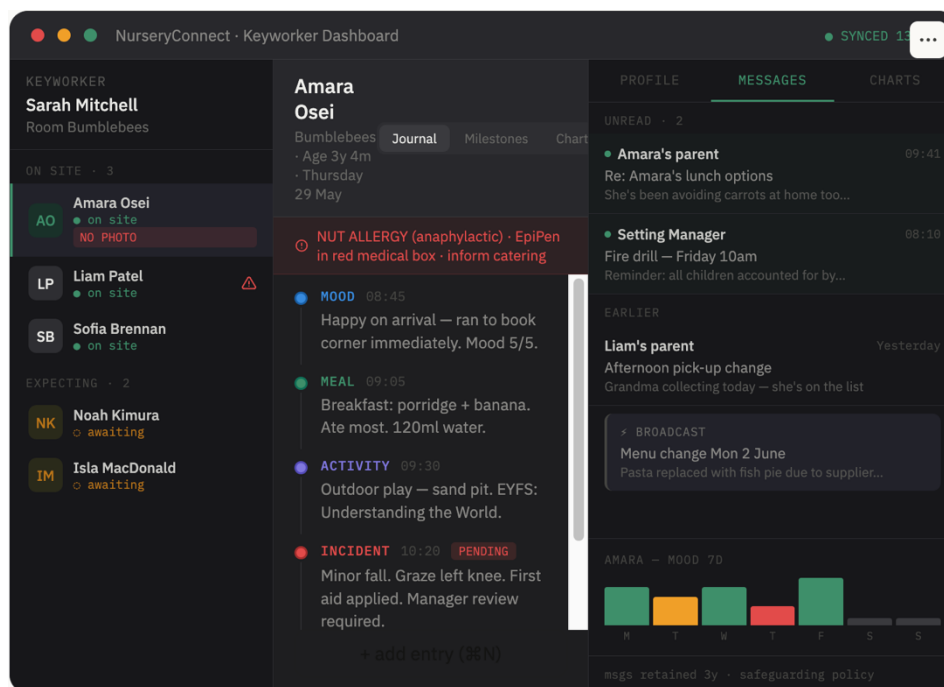
All mockup variations were generated using Claude Artifacts (`claude.ai`). The tool was prompted iteratively, with each variation responding to weaknesses identified in the previous one. Six screens are presented across the three variations below.

Variation 1 – Dark Professional Theme

Prompt summary: Generate a macOS-style dark-theme iPadOS dashboard. Three-panel layout with a dark sidebar for the children list, a center panel for the selected child's diary journal, and a right panel for messages and analytics. Use monospaced accent labels and dense information layout.

Outcome: The mockup produced a fully dark three-panel layout. The left sidebar used a very dark background with the keyworker name and room at the top, followed by ON SITE and EXPECTING child groups. Each row showed initials in a coloured circle, attendance status, and a warning icon where applicable. Selecting Amara Osei populated the centre with her journal timeline — MOOD, MEAL, ACTIVITY, and a PENDING INCIDENT entry — preceded by a prominent NUT ALLERGY warning banner. The right panel exposed Profile, Messages, and Charts tabs; the Messages tab showed a sectioned inbox (UNREAD, EARLIER) with a compact MOOD 7D sparkline anchored at the bottom.

Critical Evaluation: The information density and three-panel structure correctly used iPadOS horizontal space, and the allergy banner positioning — visible at the top of every journal interaction — was effective. However, the dark theme was judged inappropriate for a childcare environment. Nursery settings are physically light and welcoming, and a black interface would appear jarring on a device shared in the proximity of children and parents. The macOS window chrome (traffic-light buttons, title bar) was inconsistent with iPadOS conventions. The three-panel structural concept and sidebar information hierarchy were retained; the dark colour scheme was not.



[Figure 1: Mockup Variation 1 – Dark Professional Theme (Generated via Claude Artifacts)]

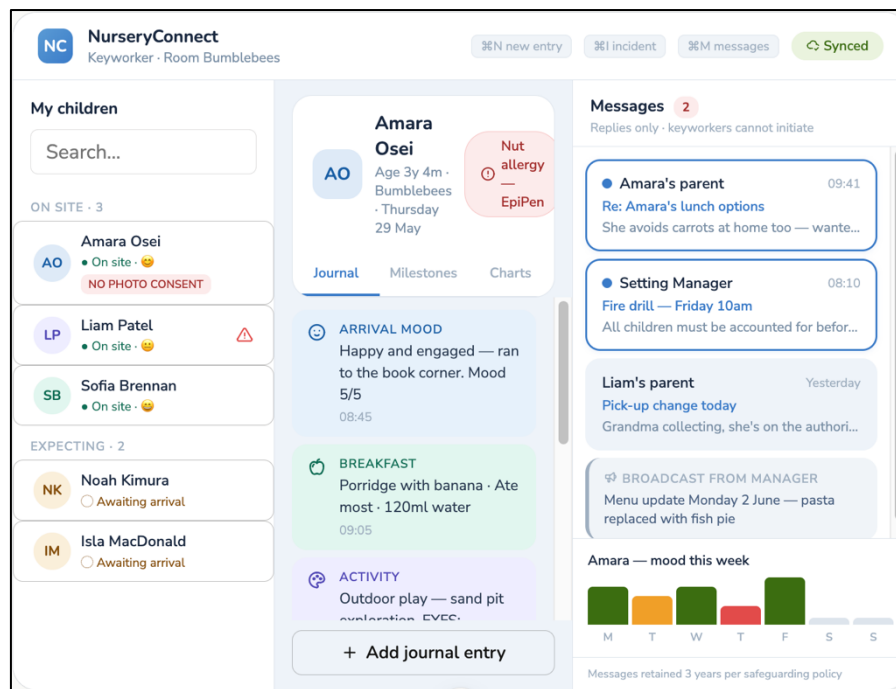
Variation 2 – Clean Light Theme with Keyboard Shortcut Toolbar

Prompt summary: Redesign as a white-background iPadOS NavigationSplitView application. Retain the three-panel layout. Show keyboard shortcuts in the toolbar (⌘N, ⌘I, ⌘M). Add an allergy badge to the selected child's profile card. Use light blue for unread message indicators.

Outcome: The mockup produced a clean white three-panel design. The top toolbar showed the NurseryConnect logo, room identifier, three keyboard shortcut buttons, and a Synced status pill. The sidebar used "My children" as the column heading with a search field and ON SITE / EXPECTING attendance groups; child rows showed pastel-circle initials, mood emoji, and warning triangles. The centre panel showed the selected child's profile card with a coral allergy badge, a Journal / Milestones / Charts segment, and a diary timeline. The right panel showed a Messages (2) header with per-thread blue unread dots, a broadcast from manager card, and a mood-this-week bar chart.

Critical Evaluation: This variation most closely conformed to iPadOS Human Interface Guidelines — NavigationSplitView was correctly applied, the search field was consistent with iPad productivity applications, and unread indicators were clear. However, the white-dominant palette was too clinical for a childcare context, resembling a medical or enterprise application

rather than a warm nursery tool. The keyboard shortcut toolbar presented ⌘N / ⌘I / ⌘M as visible UI buttons rather than CommandMenu entries, conflating two different mechanisms. Despite these issues, the structural decisions — sidebar layout, toolbar placement, allergy badge position, and mood chart integration — were carried forward into the final design.

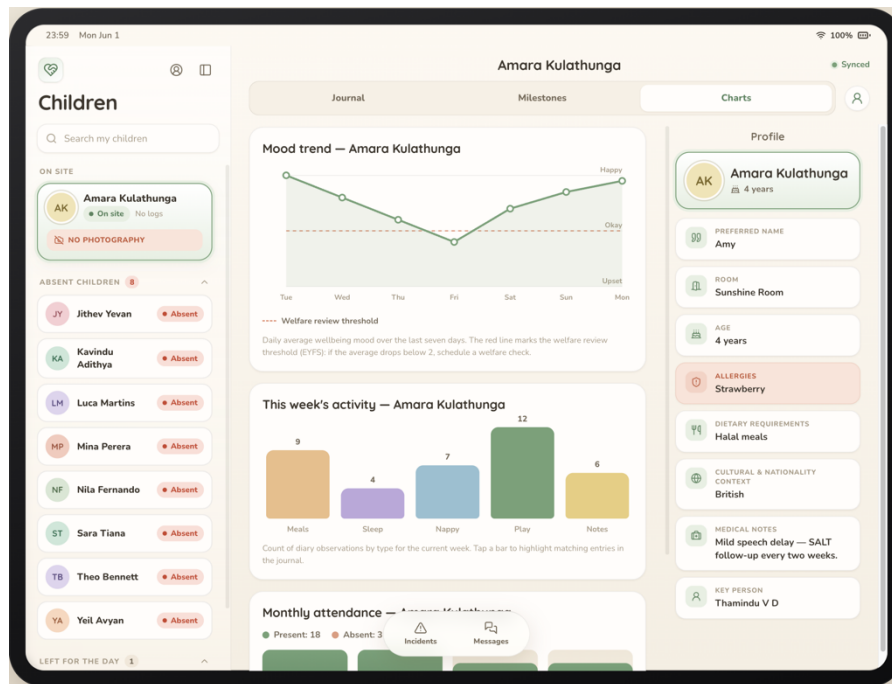


[Figure 2: Mockup Variation 2 – Clean Light Theme with Keyboard Shortcut Toolbar (Generated via Claude Artifacts)]

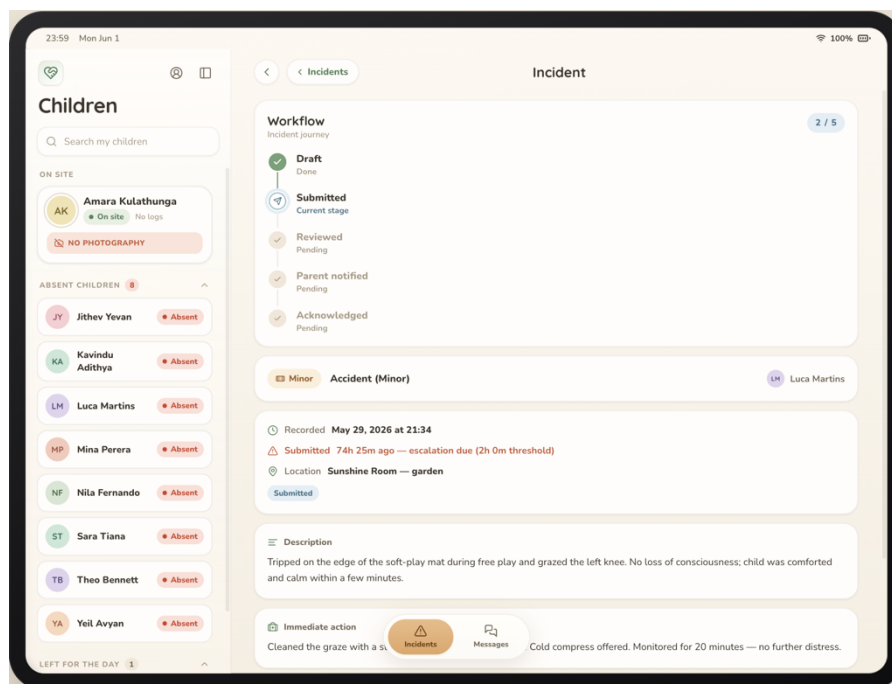
Variation 3 – Warm Beige EYFS-Aligned Theme (Selected Design)

Prompt summary: Generate a third design with a warm off-white/beige background, rounded card corners, and a muted sage-green accent. Integrate Swift Charts analytics directly into the main content panel. Show the EYFS welfare threshold as a dashed red line on the mood trend chart. Include an incident list with urgency banners and a compliance monitoring notice. Include a secure messaging view with a parent engagement sparkline and a split thread detail panel.

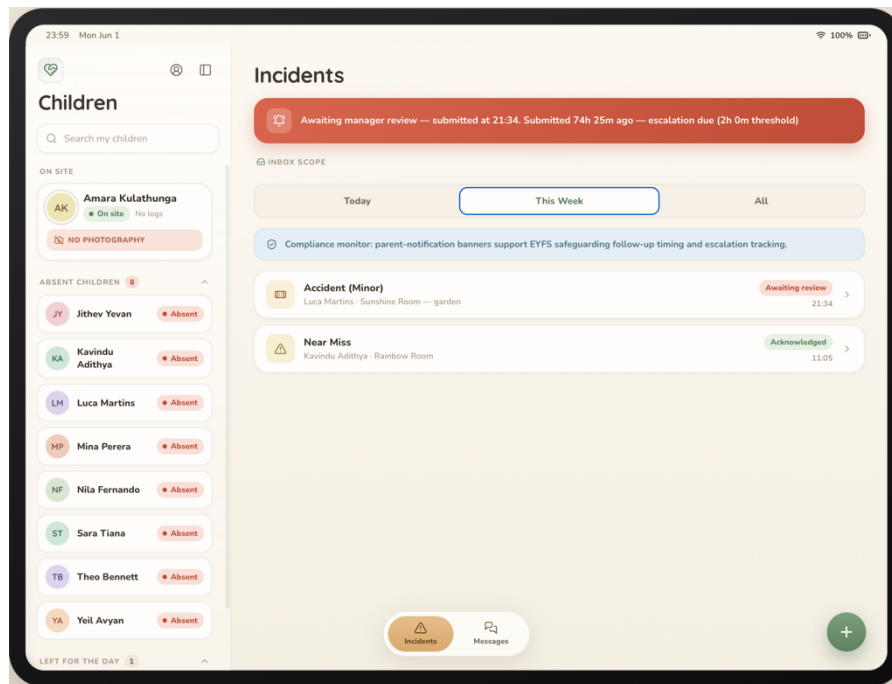
Outcome: The mockup produced four screens — Charts, Incident Detail, Incidents List, and Secure Messaging — all sharing a consistent warm beige background. The persistent left sidebar showed a "Children" column with search, an on-site section with a NO PHOTOGRAPHY warning, and a collapsible ABSENT CHILDREN section. A floating pill tab bar at the bottom replaced a full-width tab bar.



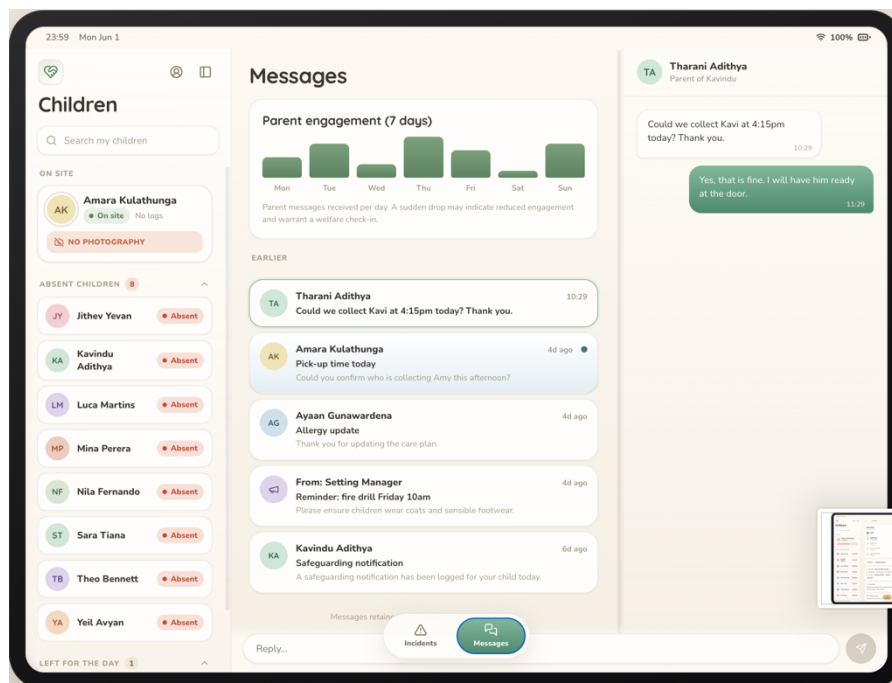
[Figure 3: Charts Screen: The Charts segment displayed a mood trend line chart with a dashed red EYFS welfare review threshold line, a tappable weekly activity bar chart, and a monthly attendance calendar grid. The right panel showed the child profile with name, room, age, allergy, dietary requirements, cultural context, medical notes, and key person]



[Figure 4: Incident Detail Screen: A five-step workflow tracker (Draft → Submitted → Reviewed → Parent Notified → Acknowledged) at step 2/5, followed by the incident record showing category badge, child name, recorded timestamp, submission time with a red escalation warning ("74h 25m ago - escalation due"), location, description, and immediate action taken]



[Figure 5: Incidents List Screen: A prominent red urgency banner at the top, a Today / This Week / All scope filter, a compliance monitor information notice, and incident rows with category icon, child name, room, status badge, and timestamp]



[Figure 6: Secure Messaging Screen: Left content panel showing a "Parent engagement (7 days)" bar sparkline, followed by the thread inbox including parent messages, a Setting Manager broadcast with a distinct icon, and older threads. Right detail panel showing the selected thread with green and white chat bubbles, a Reply field, and a "Messages retained 3 years per safeguarding policy" footer]

Critical Evaluation: This was the only variation to integrate EYFS regulatory indicators — the welfare threshold line, the escalation timer, and the safeguarding retention notice — as first-class visible elements within the normal workflow rather than buried in supplementary views. The warm beige palette was appropriate for a real nursery deployment: professional without

being clinical, and visually appropriate in a setting shared with children and parents. The floating pill tab bar correctly avoided the full-width bottom tab bar anti-pattern while remaining familiar. The only weaknesses were the absence of explicit keyboard shortcut hints and a slightly uncompact right panel profile layout, both resolved in implementation.

2.5.3 Final Design Rationale and Selection

Variation 3 was selected as the final design. The decision rested on three criteria. First, it was the only variation to surface EYFS regulatory indicators — the welfare threshold line, the escalation timer, and the safeguarding retention notice — as first-class elements in the main workflow. Second, the warm beige palette and rounded card corners were most appropriate for a real-world nursery deployment: approachable for staff of varying digital literacy and visually consistent with a childcare environment. Third, the floating pill tab bar addressed iPadOS platform conventions while remaining familiar to staff transitioning from the iPhone version.

Elements adopted from Variations 1 and 2: the three-panel `NavigationSplitView` structural concept and allergy banner positioning from Variation 1; the keyboard shortcut approach implemented as a `CommandMenu` (not UI buttons), allergy badge placement on the profile card, and mood sparkline integration from Variation 2. The dark colour scheme of Variation 1 and the white-dominant palette of Variation 2 were both rejected. Dynamic Type compliance and `.accessibilityLabel` annotations were added during implementation to extend accessibility beyond what any of the three mockups demonstrated.

2.6 Testing and Debugging

The test suite comprises 13 test files covering all major features with targeted unit tests and integration smoke tests. Tests use an in-memory `NSPersistentContainer` with fresh entity creation per test method, ensuring complete isolation and deterministic execution regardless of device state.

2.6.1 Test Suite Overview

Table 5: NurseryConnect Test Suite Coverage by Feature Area

Test File	Coverage Area	Key Assertions
MessagingViewModelTests	Reply logic, rate limiting	Parent-initiated threads allow keyworker replies; broadcast threads block replies; end-of-day summary blocked before 15:00 and on second send within the same calendar day
AttendanceViewModelTests	Attendance bucket calculation	Session weekday matching; on-site vs awaiting vs absent classification; dropoff name captured correctly
DailyDiaryViewModelTests	Entry CRUD and corrections	New entries set correct entryType; corrections create DiaryEntryCorrection with fieldName, oldValue, newValue, reason; originalSnapshotJSON populated
IncidentViewModelTests	Filtering and status workflow	Filter by status returns correct subset; severity transitions valid; isParentNotified flag toggled correctly
SyncQueueServiceTests	Offline sync queue	Pending items enqueued; retry uses exponential backoff; entity marked synced on success
MilestonePhotoPrivacyTests	Face blur pipeline	Blurred image pixel data differs from source; blurredFaceCount > 0 when faces present in test image
AuthorisedCollectorsParsingTests	Collector identity parsing	JSON round-trips without data loss; unknown collector name triggers warning flag
DiaryEntryAuditTests	Audit trail integrity	Correction chain fully reconstructable from originalSnapshotJSON; sequence of corrections ordered correctly
ChildSessionScheduleTests	Session weekday parsing	Weekday string '2,3,4,5,6' parses to Monday–Friday correctly; edge case of empty string handled
KeyworkerAttendanceBucketTests	Attendance aggregation	Present and absent counts match records; bucket boundaries correct across all session types
NurseryConnectUITests	Integration smoke tests	Application launches without crash; main navigation paths reachable; critical form presentations function

2.6.2 Edge Case Coverage and Error Handling

Several test files specifically target edge cases identified during development. The MessagingViewModelTests verify that the end-of-day summary rate limit enforces both constraints independently: the 15:00 time constraint and the one-per-child-per-day constraint cannot be bypassed by each other. The AuthorisedCollectorsParsingTests verify that a collector name does not present in the JSON authorized list triggers a safeguarding warning in the check-out sheet UI, a statutory requirement. The SyncQueueServiceTests verify exponential backoff to confirm the application does not repeatedly request a failed endpoint on every background cycle. Error handling throughout the application uses Swift typed error enumerations for sync errors, predicate construction failures, and face-detection pipeline failures, ensuring all failure paths produce meaningful diagnostic information rather than silent state inconsistencies.

2.7 Regulatory Compliance

2.7.1 GDPR (General Data Protection Regulation)

GDPR compliance is embedded in the data model and access layer from the outset rather than applied post-development. Data minimization is enforced by making optional attributes (medicalNotes, dietaryRequirements, allergies) nil when not applicable, avoiding the storage of personal data that serves no current purpose. Access scoping is enforced at the NSFFetchRequest predicate level through KeyworkerGDPRScope, ensuring a keyworker's ViewModel only ever receives data for children explicitly assigned to them. An immutable audit trail is maintained through DiaryEntryCorrection records: every field change captures the original value, new value, correction reason, and timestamp, and the full original entry state is preserved in originalSnapshotJSON to make the complete correction sequence fully reconstructable. Photo privacy is implemented through the FaceBlurEditorView pipeline: Vision detects all faces in a milestone photo, the keyworker preserves the assigned child's face, Core Image blurs all other faces before storage, and the original unblurred image is discarded immediately without persistence, consistent with the ICO's data minimization and purpose limitation principles [5].

2.7.2 EYFS (Early Years Foundation Stage)

The EYFS Statutory Framework [6] requires active welfare monitoring and clear intervention triggers. The Mood Trend chart's red dashed RuleMark at score 2 implements this requirement as a data-driven, non-intrusive visual cue embedded in the normal keyworker workflow: when the 7-day wellbeing average drops below 2, a welfare review banner is automatically surfaced in the JournalAnalyticsSummaryCard and in the visionOS mood chart overlay. The incident reporting module captures all fields mandated by the Keeping Children Safe in Education statutory guidance [7]: category, severity, location, witnesses, immediate action taken, parent notification timestamp, manager countersignature record, and RIDDOR flag. The authorized collector validation in the check-out sheet implements the safeguarding gate required to prevent child collection by individuals not on the authorized list, warning the keyworker before any collection is confirmed.

2.7.3 ICO (Information Commissioner's Office) Guidance

The application never stores biometric identifiers. The Vision face-detection operation is used exclusively for privacy protection (blurring non-consented faces) and does not result in any biometric data being persisted. All personal data access is scoped by keyworker assignment relationship, consistent with the ICO's principle of purpose limitation [5]. The absence of login and authentication screens as required by the assignment brief is compensated for through GDPR scope guards at every data access point, ensuring the application cannot expose one child's data to a keyworker assigned to a different child regardless of which device it is running on.

3 Part B – visionOS Spatial Application

The visionOS target, NurseryConnectSpatial, is a standalone spatial computing companion application that reimagines the keyworker's childcare management environment as an immersive spatial workspace. It targets visionOS 1.0+ running on Apple Vision Pro and uses an in-memory Core Data stack seeded with realistic demo data to enable full demonstration without a live backend. The application extends the Part A NurseryConnect iPadOS application into the spatial computing domain, following Apple's hub-and-spoke window architecture and spatial design principles [8].

3.1 Concept and Justification

The visionOS application targets the Setting Manager and Senior Keyworker role, nursery professionals who must simultaneously monitor multiple operational streams: attendance ratios, incident flags, transport arrival, and parent messaging. In a conventional iPad workflow, this requires repeated navigation between screens, with only one stream visible at a time. In the spatial environment, multiple focused windows can be positioned simultaneously in the user's field of view at different angles and depths, enabling a supervisor to maintain awareness of concurrent situations without switching contexts.

This is a fundamentally different interaction model from the iPad application and demonstrates genuine value from spatial computing beyond recreating flat UI in three dimensions. The 3D mood chart, for example, gives the Setting Manager a data visualization that can be physically walked around and inspected from multiple angles, a form of data engagement impossible on a flat screen. The Attendance window's real-time EYFS ratio panel provides persistent compliance monitoring in peripheral view while the Setting Manager attends to other tasks.

The choice of visionOS for Part B is further justified by the nature of childcare supervision as an inherently multi-threaded activity: a Setting Manager must maintain awareness of multiple concurrent operational streams simultaneously. Spatial computing removes the sequential bottleneck of single-screen navigation and replaces it with a spatially composed workspace where information streams coexist in physical space.

3.2 Spatial Window Architecture

The application defines seven windows registered in the App struct. Each window has a designated role, size, and window style. The `windowStyle(.plain)` windows render as flat glass panels using visionOS's `.ultraThinMaterial` aesthetic, while the Mood Chart window uses `windowStyle(.volumetric)` to render in a defined 3D volume in the user's physical space.

Table 6: NurseryConnectSpatial Window Architecture

Window ID	Style	Size	Primary Purpose
Hub	<code>.windowStyle(.plain)</code>	1000 × 700 pt	Entry point, time-of-day greeting, nursery stats summary, workspace tile grid
Keyworker Dashboard	<code>.windowStyle(.plain)</code>	980 × 720 pt	Parent message inbox and incident notification inbox
Attendance	<code>.windowStyle(.plain)</code>	1040 × 760 pt	Child check-in/check-out and live EYFS staff-to-child ratio monitor
Transport Tracker	<code>.windowStyle(.plain)</code>	1060 × 720 pt	Live nursery minibus route map, boarding manifest, and ETA indicator

Meal Plan	.windowStyle(.plain)	1100 × 720 pt	Daily meal scheduling interface for dietary and allergy management
Messaging	.windowStyle(.plain)	980 × 700 pt	Secure parent–keyworker messaging thread view
Mood Chart	.windowStyle(.volumetric)	0.48 m × 0.58 m × 0.42 m	RealityKit volumetric 3D bar chart with welfare threshold plane, drag-to-rotate

3.3 Key Spatial Features

3.3.1 Hub View

The Hub is the application's persistent entry point. It displays a contextual time-of-day greeting (Good Morning, Good Afternoon, or Good Evening based on system time), four nursery-wide statistics in prominent tiles (enrolled children, currently on-site, active incidents, unread messages), and a 3-column grid of workspace tiles. Each workspace tile uses `.ultraThinMaterial` for the visionOS glass-panel visual language and opens its associated window via `openWindow(id:)` when tapped. The Hub is designed to remain visible while other workspace windows are active, serving as the persistent anchor in the spatial layout, consistent with the approach demonstrated in visionOS apps such as Freeform and Mindfulness.

3.3.2 Spatial Attendance View with EYFS Ratio Monitoring

The Attendance window is the most fully featured screen in the spatial application. It presents a `LazyVGrid` of child cards with adaptive column widths (200–260 pt), each showing a circular initials avatar, name, room assignment, and a color-coded attendance status badge. A persistent real-time ratios panel on the right side of the window displays live EYFS compliance status for three age groups: 1:3 (under 2s), 1:4 (2-year-olds), and 1:8 (3–5 years). Each ratio row is colored green when compliant and red when a breach is detected, providing an immediate visual safeguard against statutory ratio violations.

Attendance check-in and check-out are presented as `.sheet(item:)` sheets, using the optional record as the presentation gate to prevent the race condition identified during development (see Section 5.3). The check-in sheet collects the dropper-off name, mood level, and freeform notes. The check-out sheet validates the stated collector against the `authorisedCollectors` JSON field and displays a clear warning banner if the collector is not on the approved list, implementing the statutory safeguarding gate for child collection.

3.3.3 Three-Dimensional Mood Chart (RealityKit)

The Mood Chart window is the most technically distinctive feature of the visionOS application and its most compelling spatial computing demonstration. It opens as a volumetric window measuring 0.48 m × 0.58 m × 0.42 m, a defined physical volume in the user's real-world space. Inside this volume, seven `ModelEntity` bars are placed along the X axis using RealityKit's Y-up, meter-unit coordinate system. Each bar's height is normalized to a maximum `barHeight` of 0.11 m and colored according to the mood score (red for scores 1–2, amber for 3, green for 4–5). A flat `ModelEntity` plane at `Y = -0.04` acts as the EYFS welfare threshold marker.

```
// MoodChartVolumeView.swift - RealityKit bar construction
let mesh = MeshResource.generateBox(
    size: SIMD3<Float>(barWidth, barHeight, barWidth),
    cornerRadius: 0.004
```

```

)
let material = SimpleMaterial(color: barColor, isMetallic: false)
let barEntity = ModelEntity(mesh: mesh, materials: [material])
barEntity.position = SIMD3<Float>(xOffset, floorY + barHeight / 2, 0)
root.addChild(barEntity)

```

The chart supports drag-to-rotate gesture interaction, allowing the Setting Manager to inspect bar heights from multiple viewing angles. Day labels (Monday through Sunday) are rendered as overlaid SwiftUI Text views using RealityView's attachments API, combining RealityKit geometry with SwiftUI typography in the same spatial scene. This gives the Setting Manager a data visualization that can be physically walked around, a fundamentally new form of data engagement enabled by spatial computing that cannot be replicated on any flat-screen device.

3.3.4 Transport Tracker

The Transport Tracker simulates a nursery minibus route in real time using animated MapKit annotations and a boarding manifest. A MapKit map panel displays the current van position as an animated annotation updated every three seconds via `Timer.publish(every: 3)`, along with the nursery location pin and planned stop pins. A boarding manifest panel lists the full route in order with a checked or pending boarding status for each child. A status bar summarizes boarding progress (X of Y children boarded) and the estimated arrival time. This feature addresses a genuine operational need: a Setting Manager can position the Transport Tracker window in peripheral spatial view while attending to front-desk duties, maintaining awareness of transport status without requiring repeated screen navigation.

3.3.5 Secure Messaging

The Messaging window in the spatial application provides the same Core Data-backed parent-keyworker messaging inbox as the iPadOS application, adapted to the visionOS visual language. Message threads are displayed with `.ultraThinMaterial` card backgrounds. The chat bubble thread detail uses the same `MessagingViewModel` as the iPadOS target, demonstrating code-sharing between platforms. Reply restrictions and end-of-day summary rate limiting from the iPadOS implementation are fully present in the spatial version.

3.4 Spatial UX Design

The application follows Apple's visionOS spatial design principles [8] throughout. Three specific applications of these principles are noteworthy. First, all windows are sized to coexist in the user's field of view without spatial collision when opened in sequence from the Hub, reflecting the principle that spatial apps should be spatially composed rather than spatially competing. Second, `.ultraThinMaterial` is used for all panel backgrounds, ensuring windows feel like glass panels floating in the physical environment rather than opaque screen elements dropped into space, consistent with visionOS's core visual language. Third, the Hub persists as an anchor while subordinate windows are open, mirroring the recommended hub-and-spoke pattern for spatial productivity applications described in Apple's spatial design documentation [8].

3.5 Simulator versus Device Code Paths

RealityKit volumetric features, including `MeshResource.generateBox` and the volumetric `windowStyle`, are not available in the visionOS simulator architecture. To enable building,

running, and demonstrating the application in the simulator without compilation failures, the mood chart view uses a conditional compilation guard:

```
#if arch(simulator)
    MoodChartBarStripView(...)    // 2D bar chart with perspective(1000)
    transform
#else
    RealityViewContainer(...)    // Full volumetric 3D RealityKit chart on
    device
#endif
```

The 2D fallback `MoodChartBarStripView` applies a `.perspective(1000)` transform to a standard SwiftUI bar chart, providing a visual approximation of depth in the simulator. All other windows function identically in simulator and on device, as they use only `flat windowStyle(.plain)` and standard SwiftUI views.

4 AI Usage Documentation

This section documents all AI interactions undertaken during the development of this assignment. AI tools were used in four phases: architecture decision-making, feature implementation, Apple HIG auditing, and visual bug diagnosis. All code generated by AI tools was reviewed, tested in the simulator, and in several cases rejected or substantially modified before use.

4.1 UI Mockup Generation – Claude Artifacts

All three UI mockup variations described in Section 2.5.2 were generated using Claude Artifacts at claude.ai. The complete prompts and outcomes are documented in Table 9 below. Mockup screenshots are presented in Figures 1–3.

Table 8: Complete AI Mockup Generation Prompt and Outcome Log

Variation	Tool	Full Prompt (Summarized)	AI Response Summary	Design Decision
1 – Minimalist Grid	Claude Artifacts (claude.ai)	Generate an iPadOS SwiftUI mockup for a childcare keyworker app. Include child list, diary entries, attendance status. Use clean minimal white design with card grid and bottom tab bar.	Produced a white-background card grid layout with circular avatars and bottom tab bar with five sections. Clean and readable but structured as an iPhone-first design.	Rejected as primary layout. Bottom tab bar is not appropriate for regular-class iPadOS. Card grid does not leverage iPad horizontal space. Aesthetic quality retained as reference.
2 – Sidebar Professional	Claude Artifacts (claude.ai)	Redesign as NavigationSplitView layout with persistent sidebar. Child rows with attendance badge and diary dot. Deep teal accent. Add detail panel for message and incident content.	Produced a NavigationSplitView mock with compact sidebar rows, color-coded attendance indicators, and a teal accent system. Correct iPad layout pattern.	Layout adopted. Color palette revised, deep teal judged too clinical for childcare. Sidebar structure, column widths, and compact row card design carried into implementation.
3 – EYFS Warm Palette	Claude Artifacts (claude.ai)	Third variation with warm EYFS color palette (coral, sage, sky-blue), rounded corners, large tap targets, high-contrast text labels, Dock shortcut row at bottom.	Produced a warm, rounded design with coral-green accents, oversized tap targets, and a Dock-style action row. Accessibility compliance clearly demonstrated.	Color palette, rounded corners, and accessibility decisions adopted. Dock pattern rejected, unfamiliar iPadOS paradigm. Warmth reinterpreted through nursery green accent and rounded card corners in final design.

4.2 Code Generation – Claude (claude.ai)

4.2.1 Phase 1 – Architecture Decisions

High-level architectural questions were posed to Claude covering three topics: the split-view layout approach, the messaging data model, and GDPR pattern design.

Prompt: I am building an iPadOS childcare management app with a three-panel workspace (sidebar, content, detail). Should I use NavigationSplitView or a custom ZStack-based layout? What are the trade-offs?

AI Response (summarized): NavigationSplitView is strongly preferred because it handles sidebar show/hide animation natively, integrates with the system keyboard shortcut Ctrl+Shift+S, provides the columnVisibility API for programmatic control, and receives future Apple enhancements automatically. The ZStack approach requires manually reimplementing these behaviors and is unlikely to remain consistent across iPadOS updates.

Action: NavigationSplitView adopted. The custom ZStack prototype was discarded. The recommendation was cross-validated against WWDC24 Session 10147 [1] before implementation.

4.2.2 Phase 2 – Performance and Memory Optimization

Prompt (Core Data Fetch Request Overhead): The AnalyticsDataService runs multiple @FetchRequest calls on the same view. The app feels sluggish when switching between children in the sidebar. How do I diagnose and reduce Core Data fetch overhead in a SwiftUI context?

AI Response (summarized): The likely cause is unbounded fetch requests with no fetchLimit or batchSize, combined with fetches firing on every view re-render. Recommendations: add batchSize: 20 to all list fetches, add fetchLimit where only a fixed window of records is needed (e.g., last 7 days), use NSSortDescriptor with an indexed attribute so Core Data can short-circuit the sort, and move @FetchRequest properties to dedicated child views rather than a single parent so only the relevant fetch re-fires when state changes.

Action: batchSize and fetchLimit were added to the mood trend and message sparkline fetches. The analytics dashboard was refactored, so each chart view owns its own @FetchRequest rather than deriving from a shared parent fetch. Sidebar switching latency was eliminated after these changes.

Prompt (FaceBlurEditorView Memory Pressure): The app crashes on older iPad models when a keyworker selects a high-resolution photo for a milestone entry. The Vision face detection and Core Image blur pipeline is processing the full-resolution image. How should I cap memory usage without degrading blur quality?

AI Response (summarized): Downsample the image before passing it to VNDetectFaceRectanglesRequest. A maximum input dimension of 1600 pt is sufficient for face detection accuracy; Vision's bounding boxes are normalized so they scale correctly back to any output size. Also normalize EXIF orientation before downsampling — orientation-mismatched bounding boxes are a common bug when skipping this step. Apply blur to the downsampled image, not the original; the difference is imperceptible at milestone photo display sizes.

Action: A pre-processing step was added to FaceBlurEditorView that caps the input image at 1600 pt and applies UIImage orientation normalisation before the Vision request. Memory

pressure on low-end iPad models was resolved. The blurredFaceCount stored in Core Data reflects detections on the downsampled image, which was verified to match the original in the MilestonePhotoPrivacyTests suite.

Prompt (SwiftUI Unnecessary Re-renders): Charts in ChildAnalyticsDashboardView are re-animating every time I update any unrelated state in the parent view. How do I prevent this?

AI Response (summarized): The .animation modifier without a value: parameter responds to any state change on the view, not just the relevant one. Replace .animation(.spring()) with .animation(.spring(), value: specificStateProperty) so the animation only triggers when that property changes. For charts, also ensure @FetchRequest properties are on the chart view itself rather than a parent — a fetch result change on a parent triggers a full subtree re-render.

Action: All .animation calls were audited and converted to the value-scoped form, as referenced in WWDC23 Session 10156 [2]. Unrelated state changes in KeyworkerIPadCoordinator no longer cause chart animations to re-fire.

4.2.3 Phase 3 – Apple HIG Audit

Prompt: Act as a senior iOS engineer auditing the NurseryConnect application for Apple Human Interface Guidelines compliance. Identify the 10 highest-priority violations and improvements, with justification for each.

AI Response (summarized): The audit returned a prioritized list of 10 items: convert custom button styles to .borderedProminent/.bordered, migrate fonts to Dynamic Type semantic styles, replace custom empty states with ContentUnavailableView, use presentationDetents on all sheets, add .hoverEffect to sidebar rows, enforce a single accent color, add pull-to-refresh to lists, apply .searchable to child list, add keyboard shortcuts, and use correct presentation modifiers.

Action: All 10 identified items were implemented. The Dynamic Type migration in particular required visual review of every screen (see Section 5.6).

4.2.4 Phase 4 – Bug Diagnosis

Prompt (visionOS Sheet Race Condition): My visionOS attendance check-in sheets are presenting empty. I have @State var selectedRecord: NCAttendanceRecord? and @State var showsCheckInSheet: Bool. Attaching code and screenshot. What is causing this?

AI Diagnosis: This is a classic SwiftUI presentation race condition. The sheet is presented when showsCheckInSheet becomes true, but selectedRecord may not yet have propagated. Replace both state variables with .sheet(item: \$checkInRecord) where checkInRecord is an optional NCAttendanceRecord. The optional itself serves as the gate.

Action: .sheet(item:) pattern adopted throughout both targets. Race condition eliminated completely in testing.

4.2.5 Rejected AI Outputs and Critical Review

Several AI-generated code patterns were identified as incorrect and rejected during the mandatory review phase:

- A recursive ButtonStyle.makeBody implementation that called .buttonStyle() on its own content internally. This is invalid in SwiftUI because button styles cannot call other button styles recursively. The style was rewritten as an opacity-only pass-through.

- An initialization call `KeyworkerChildSummary(from: child)` using an initializer that did not exist in the codebase. Accepted without verification, this would have caused a compile-time error. The correct construction path was identified from the model definition and used instead.
- The `summaryPill(label:count:color:Any)` function signature using an `Any`-typed color parameter. This caused cascading Swift type inference failures in surrounding `@ViewBuilder` closures, producing misleading compiler error messages pointing to unrelated lines. The function was rewritten with an explicit `Color` parameter and the `ContentUnavailableView` extracted into a separate computed property.

These cases demonstrate that AI-assisted development requires a higher critical review standard, not a lower one. AI tools generate syntactically plausible code very quickly, but plausibility is not correctness. All generated code must be understood, tested, and independently verified before integration. The developer's domain expertise, architectural judgment, and code review skills remain essential and non-delegable.

5 Challenges Faced

5.1 SwiftUI Type Inference with Generic Closures

The most time-consuming compiler issue arose in `ChildMonthlyAttendanceChart` following the migration from `BarMark` to the custom dot-grid calendar. The `summaryPill` helper function had been given an `Any`-typed color parameter during an early prototype draft. Swift's bidirectional type inference in `@ViewBuilder` result builders failed to resolve the `Group` branch types, producing spurious 'cannot convert to `TableColumn`' errors pointing to lines that were not the source of the problem. The resolution required giving `summaryPill` an explicit `Color` parameter and extracting the `ContentUnavailableView` into a separate computed property, reducing the complexity of the `@ViewBuilder` closure to a level Swift's type checker could resolve.

Lesson: Swift's type inference error messages in result builders frequently identify the symptom (a line where inference failed) rather than the cause (a type ambiguity several lines earlier). When confusing result-builder errors appear, the most productive approach is simplifying the closure structure and adding explicit type annotations rather than attempting to diagnose the reported error location.

5.2 RealityKit Volumetric Features on the visionOS Simulator

RealityKit volumetric features, including `MeshResource.generateBox`, volumetric window styles, and the RealityView coordinate system, are not available on the visionOS simulator architecture. Early builds failed to compile because these APIs do not exist on the simulator target. The conditional compilation guard `#if arch(simulator)` was added to provide a 2D perspective-transform fallback for simulator builds while preserving the full RealityKit implementation for the device. This required maintaining two rendering paths for the mood chart view, increasing the code surface area, but was essential for enabling simulator-based development and demonstration.

5.3 Sheet Race Condition in visionOS Attendance View

The attendance check-in sheets in `SpatialAttendanceView` initially presented with empty content on an intermittent basis, roughly one in three taps. Investigation identified a race condition between `@State` var `selectedRecord` and `@State` var `showsCheckInSheet`: SwiftUI was presenting the sheet when `showsCheckInSheet` became true before the update to `selectedRecord` had fully propagated through the state graph, resulting in the sheet reading nil for its required record. The fix was replacing both state variables with `.sheet(item: $checkInRecord)`, where the optional `checkInRecord` itself is the presentation gate. Setting it to a non-nil value both triggers the presentation and provides the data simultaneously, eliminating the race entirely. This pattern is now applied throughout both targets as the default approach for all item-driven sheet presentations.

5.4 iPad Split-View Column Alignment

The minimized sidebar rail, 62 pt wide when the `NavigationSplitView` sidebar is fully collapsed, required precise coordination between the column width API, the `splitColumnVisibility` state, and two custom `Environment` values (`usesFloatingTabBarShell` and `keyworkerSidebarHidden`). Achieving a smooth reveal and hide animation consistent

across all supported iPad sizes (iPad mini, iPad Air, iPad Pro 11-inch, iPad Pro 13-inch) took multiple development iterations. The `NavigationSplitView` column width API does not expose direct access to the animation curve used for column transitions, requiring careful timing alignment with the column visibility changes.

5.5 GDPR Predicate Construction

NSPredicate format strings for cross-entity relationship filtering are error-prone: a key-path typo produces a silent runtime crash with no compile-time diagnostic. This was encountered during development of the diary entry fetch predicates that filter through a relationship chain. The resolution was centralizing all predicates in `AnalyticsDataService` as static functions and writing a dedicated unit test for each predicate. Key-path typos that previously caused silent runtime crashes now produce immediate test failures, giving the developer both a clear error message and a known-good predicate as the baseline.

5.6 Dynamic Type Migration and Custom Fonts

The original Assignment 1 codebase used `.system(.title2, design: .rounded)` throughout for stylistic purposes. This font specification is not a semantic style and does not participate in Dynamic Type, text does not scale when the user increases their preferred accessibility font size in System Settings. The migration to semantic styles (`.title2.weight(.semibold)`) resolved this accessibility failure but required comprehensive visual review of every screen. Semantic styles have slightly different baseline metrics than their explicitly-sized equivalents, and several layouts that used fixed spacing values or absolute frame heights broke when the font metrics changed, requiring spacing adjustments across more than 12 views.

6 Learning Reflection

6.1 Apple Developer Resources Consulted

The following Apple developer resources directly informed implementation decisions and are cited throughout this report:

6.1.1 WWDC23 – Explore SwiftUI Animation (Session 10156) [9]

This session clarified the critical distinction between implicit animations (triggered by any state change) and explicit, value-scoped animations. It informed the use of `.animation(.spring(response: 0.35, dampingFraction: 0.85), value: viewModel.filter)` on the incident list segmented picker, without the `value: parameter`, unrelated view state changes trigger the same animation, producing unexpected visual artefacts. The session also explained how `@FetchRequest` interacts with `withAnimation: Core Data` changes animate through the SwiftUI update cycle when `FetchRequest` is initialized with `animation: .default`, which is why all chart view `@FetchRequest` properties include this parameter.

6.1.2 WWDC24 – Elevate Your Tab and Sidebar Experience in iPadOS (Session 10147) [1]

This was the primary reference for the iPad split-view architecture. The session demonstrated `NavigationSplitView` column visibility management, programmatic sidebar collapse to a rail, and the importance of the `columnVisibility` binding for seamless sidebar state control. The minimized 62 pt avatar rail in `KeyworkerIPadShellView` was directly inspired by the compact sidebar pattern shown in this session. The session also discussed `toolbar(.hidden, for: .navigationBar)` and custom toolbar placements, both applied in `ChildrenSidebarView` for the profile button and sidebar toggle.

6.1.3 WWDC23 – What's New in Swift Charts (Session 10037) [3]

This session introduced the `RuleMark` API for reference lines and the `chartOverlay` modifier for custom interaction handling. Before watching this session, the EYFS welfare threshold line in `MoodTrendChart` was implemented as a `Rectangle` overlay in a `ZStack`. This approach did not participate in the chart's coordinate system and drifted on different screen sizes and orientations. Replacing it with `RuleMark(y: .value("Threshold", 2))` placed the threshold line precisely in data coordinates, guaranteeing correct alignment regardless of chart dimensions. The `chartOverlay` API informed the tap gesture implementation in `WeeklyActivityDistributionChart` that reads bar data values from the chart's proxy to drive diary timeline filtering.

6.1.4 Develop in Swift – Persistence (Apple Developer Tutorials) [2]

The offline-first sync architecture was informed by Apple's recommended Core Data persistence pattern: write locally first, sync asynchronously. The `SyncQueueService` with per-entity `syncState` flags follows this model directly. The tutorial's discussion of `NSPersistentCloudKitContainer` clarified why test targets must use `NSPersistentContainer` with an in-memory store, CloudKit sync integration can interfere with test isolation, which shaped the test infrastructure design described in Section 2.6.

6.1.5 Spatial Design for visionOS – Apple Developer Documentation [8]

Apple's spatial design principles shaped the hub-and-spoke window architecture. The guidance that spatial apps should feel as if they belong in the space, not that flat screens have simply been relocated into it, directly informed the decision to design multiple focused, complementary windows rather than placing the entire iPad application in a single large panel. The coordinate system documentation for RealityView (Y-up, center-origin, meters) was essential for correct bar positioning in the 3D mood chart and for placing the welfare threshold plane at the correct relative depth.

6.2 Personal Reflection

This assignment demonstrated that modern iOS and spatial computing development in 2026 requires deep familiarity with system capabilities as much as raw coding skill. The most meaningful learning moments were:

- The `.sheet(item:)` pattern eliminates an entire class of presentation race conditions by making the optional binding itself the gate. This is now the default approach for all item-driven sheet presentations in any future project.
- Swift type inference failures in `@ViewBuilder` result builders rarely point to the actual line of the error. Simplifying closure structure and adding explicit type annotations resolves them more reliably than attempting to diagnose the reported error location.
- GDPR compliance must be embedded in the data model and access layer from the first day of design. Predicate-level scoping in `NSFetchRequest` is a fundamentally stronger privacy guarantee than post-fetch filtering, which cannot be verified by unit tests and can expose data through subtle coding errors.
- AI-assisted development raises the required critical standard rather than lowering it. The AI generates plausible-looking code at high speed, but plausible is not correct. Every generated output must be understood, tested, and critically evaluated. The developer's architectural knowledge, code review judgment, and understanding of Swift's type system remain essential and cannot be outsourced to the AI.
- Spatial computing requires a genuinely different mental model from screen-based development: Y is up, distances are in meters, and depth is a real layout dimension with real-world consequences. The 3D mood chart required unlearning flat-screen layout instincts entirely. The result, a data visualization the user can physically walk around, was the most compelling demonstration of what spatial computing can uniquely offer compared to any flat-screen alternative.

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